Comprehensive Road Scene Understanding for Autonomous Driving: Semantic Segmentation to Anomaly Detection with EoMT

Omer Ozkan Ceylin Afyoncu Franck Hernold Kewo Kameni
Politecnico di Torino
Student IDs: s360761, s346382, s359890
{s360761, s346382, s359890}@studenti.polito.it

Abstract

Semantic and anomaly segmentation is critical for autonomous driving. But predefined-class models fail on the unknown objects, and it creates danger in the real world scenarios. We evaluate EoMT (encoder-only mask transformer with DINOv2 backbone) on semantic segmentation, and we compare COCO-trained EoMT (X% mIoU) with Cityscapes-trained EoMT (X% mIoU) on Cityscapes validation dataset. We fine-tune the COCO-EoMT on Cityscapes which reaches to X% mIoU. For anomaly segmentation, we compare ERFNet and EoMT models using four post-hoc scoring methods which are MSP, MaxLogit, MaxEntropy, and RbA on 5 different datasets. Our findings show that EoMT-Cityscapes with MSP reaches X% AuPRC on RoadAnomaly21, and RbA helps lowering false positive rates. Our code is available at `https://github.com/omerzkan/eomt-anomaly-segmentation.git`.

1. Introduction

Semantic segmentation analyzes every pixel and classifies each one into its related class. A vehicle needs to know where pedestrians are and where the road is; without pixel-level analysis, this is difficult. The Cityscapes dataset is a standard benchmark for this type of task. It includes 5000 photos of German streets. Each photo is manually labeled with 19 classes such as road, sidewalk, building, wall, pole, and traffic light. ERFNet [8] is a traditional pixel-based approach. It uses a CNN (convolutional neural network) with an encoder-decoder architecture, and classifies each pixel independently.

MaskFormer [3] changed this approach by predicting N binary masks and N class labels, then combining them. In this way, the model handles both semantic and panoptic segmentation. Mask2Former [4] improved MaskFormer by adding masked cross-attention, so the model only looks at the image region where each mask is predicted. EoMT [5] simplified this further. It uses only the DINOv2 [7] encoder and removes the pixel decoder entirely.

The problem is that all these models are closed-set. They were trained on exactly 19 classes, and during training they did not see anything other than what they were trained on. However, the model must assign objects it has never seen to one of its known classes. This is dangerous: if the model assigns an animal or fence as road, the vehicle tries to drive through it. The softmax function forces all probabilities to sum to 1, so even for something the model has never seen, it produces a confident prediction. This is called overconfidence on out-of-distribution (OOD) data. We used four post-hoc scoring methods (MSP, MaxLogit, MaxEntropy, RbA [6]) with temperature scaling, and evaluate them on benchmarks from SMIYC [2] and Fishyscapes [1].

We evaluated EoMT on semantic segmentation and compared two differently trained models (COCO-EoMT vs Cityscapes-EoMT) to show the importance of fine-tuning. Then we fine-tuned the COCO-EoMT on Cityscapes, and the model achieved 78.85% mIoU. Next, we compare the pixel-based and mask-based models using post-hoc anomaly detection methods across 5 benchmarks.

2. Methodology

2.1. Semantic Segmentation

Semantic segmentation was our first task, and we used EoMT [5], which is a mask-based architecture built on a DINOv2-pretrained Vision Transformer [7]. Mask classification predicts a fixed set of K masks with a class label for each mask. This idea was introduced by MaskFormer [3] and improved by Mask2Former [4]. EoMT removes the convolutional adapter and the pixel decoder, simplifying the mask-based pipeline. It only uses the pretrained ViT to extract features.

We compared two EoMT models using two different checkpoints: EoMT-COCO and EoMT-Cityscapes. A checkpoint contains the model weights learned from a specific dataset, so the same architecture behaves differently depending on which checkpoint is loaded. EoMT-COCO is trained on the COCO dataset, a large general-purpose dataset with 118k training images covering daily scenes such as streets, kitchens, beaches, and animals. COCO has 133 class labels: 80 "thing" classes such as person, car, and dog, and 53 "stuff" classes such as sky, grass, wall, and sand. This makes the model strong on a wide range of visual objects, but it is not road-specific.

The second model is EoMT-Cityscapes, trained on the Cityscapes dataset. Cityscapes is a smaller dataset (around 3k training images) of road and driving scenes from German cities. It has 19 class labels specifically chosen for autonomous driving, such as road, sidewalk, building, pole, traffic light, traffic sign, person, rider, car, and bicycle. It is road-specific but not as broad as COCO.

We evaluated both models on the Cityscapes validation set, which contains 500 images. The goal was to understand how much a general model loses compared to a specialized one, and whether fine-tuning can close the gap. To do this, we had to solve one problem: EoMT-Cityscapes has 19 output channels matching the 19 Cityscapes classes, while EoMT-COCO has 133 output channels matching the 133 COCO classes. We had to translate the COCO predictions into the Cityscapes label space before evaluation.

We built a `coco_to_cityscapes.py` mapping that works in three steps:

- **Direct name match:** obvious matches found by string comparison. For example, COCO "car" maps directly to Cityscapes "car", and COCO "person" to Cityscapes "person".
- **Synonyms:** some classes are named differently in the two datasets. We built a synonym dictionary to handle these cases. For example, COCO calls sidewalks "pavement-merged".
- **Ignored:** COCO labels with no Cityscapes equivalent are mapped to the ignore index 255, which excludes them from evaluation. For example, if EoMT-COCO predicts "dog", that pixel is ignored.

For each of the 19 Cityscapes classes we compute the Intersection-over-Union (IoU):

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{1}$$

The mean IoU (mIoU) is the average over the 19 classes. It treats every class equally, so rare classes count as much as common ones.

2.2. Fine-Tuning

In fine-tuning, we take a model which is already trained on a dataset and continue training on a new dataset while keeping the useful information from previous trainings. We chose fine-tuning because EoMT-COCO is broad and powerful, but not road-specific. We wanted to see if we could adapt it to the Cityscapes domain without losing its general knowledge.

When fine-tuning EoMT-COCO, we dropped the classification head because COCO's classification head produces 133 logits per query, one per COCO class. We replaced it with a new head that has 19 outputs which is one per Cityscapes class. The rest of the weights are kept from the EoMT-COCO. For efficiency, we froze the DINOv2 backbone and trained only the queries, mask module, and the new classification head. We also use Automatic Mixed Precision (AMP). Instead of FP32, we use mixed-precision FP16. For the training setup we used 20 epochs, $K = 200$ queries, image size 640×640. Training was done on a single NVIDIA L4 GPU via Google Colab Pro. For the loss function we used standard Mask2Former [4] loss. It combines Hungarian matching between predicted and ground-truth masks, with cross-entropy on classes and Dice + binary cross-entropy on masks. We use AdamW with a learning rate of 1×10^{-4}.

2.3. Anomaly Segmentation

We evaluate ERFNet and EoMT under anomaly segmentation. Anomaly segmentation detects pixels belonging to objects which are not found in the training class set. We can say that these pixels are out-of-distribution. For evaluation, we use post-hoc methods which use model's class logits, do not require retraining, and calculate anomaly score per pixel. Both ERFNet (pixel-based) and EoMT (mask-based) produce class logits, and this allows us to use the same scoring functions for both models. For evaluating the models, we consider four scoring methods.

Post-hoc anomaly detection methods.

- **MSP (Maximum Softmax Probability):** uses the highest softmax probability as an uncertainty indicator.

$$\text{score}_{\text{MSP}}(\mathbf{x}) = 1 - \max_c p_c(\mathbf{x}) \tag{2}$$

- **MaxLogit:** uses the raw logits before softmax, because they preserve the magnitude of the model's confidence.

$$\text{score}_{\text{MaxLogit}}(\mathbf{x}) = -\max_c L_c(\mathbf{x}) \tag{3}$$

- **MaxEntropy:** if the probability is spread across many classes (high entropy), the model is uncertain and the pixel is a possible anomaly. If it puts all probability on one class (low entropy), the model is confident.

$$\text{score}_{\text{Entropy}}(\mathbf{x}) = -\sum_{c=1}^{C} p_c(\mathbf{x}) \log p_c(\mathbf{x}) \tag{4}$$

- **RbA (Reweighted by Area) [6]:** only for mask-based models. It does not work with ERFNet [8]. RbA flags pixels that are not strongly claimed by any mask. RbA relies on an independence assumption: in a mask architecture, each class behaves like a one-vs-all classifier, so class probabilities do not need to sum to 1. In a pixel-wise classifier like ERFNet, the class probabilities must sum to 1. The independence assumption is violated, and RbA doesn't have the same OoD-detection meaning. That's why we don't apply RbA to ERFNet.

$$\text{score}_{\text{RbA}}(\mathbf{x}) = -\sum_{c=1}^{C} \tanh(L_c(\mathbf{x})) \tag{5}$$

Temperature scaling. It is a calibration technique that can shift recall and precision in anomaly detection. It divides logits by T before softmax. $T > 1$ makes probabilities more uncertain, $T < 1$ makes them more confident. We tested $T \in \{0.5, 0.75, 1.0, 1.1, 1.5, 2.0\}$ on MSP, MaxEntropy, and RbA. We didn't test on MaxLogit, which doesn't use softmax.

Datasets and metrics. As evaluation benchmarks, we use a smaller subsets of these five datasets which are provided by course instructors. These are:

- **RoadAnomaly21 (from SMIYC [2]):** 10 images, large unknown objects in diverse environments.
- **RoadObstacle21 (from SMIYC [2]):** 30 images, small unknown objects on the road.
- **Fishyscapes Lost&Found [1]:** 100 validation images of real lost-cargo objects.
- **Fishyscapes Static [1]:** 30 images, synthetic anomalies on the road scenes.
- **RoadAnomaly (from SMIYC [2]):** 60 images, earlier version of SMIYC's anomaly track.

We use two metrics:

- **AuPRC** (Area under the Precision-Recall Curve): Evaluates pixel-level separability. Higher values mean better discrimination of rare anomalies.
- **FPR@95** (False Positive Rate at 95% True Positive Rate): Measures false positive rate when the true positive rate is fixed at 95%. It is a safety critical metric. Lower is better.

3. Results

We evaluated closed-set semantic segmentation on Cityscapes validation datasets, and anomaly segmentation on 5 benchmarks. Our findings include that mask-based EoMT model outperforms the pixel-based ERFNet on both tasks. Another interesting point that fine-tuning on COCO-EoMT improves mIoU result but degrades anomaly performance. As last, temperature scaling has limited effect on the anomaly metrics for this task.

Table 1. Closed-set semantic segmentation results on `Cityscapes validation dataset` (mIoU %). EoMT-COCO is evaluated zero-shot with a class-mapping from COCO to Cityscapes. EoMT-Fine-tuned originally come from EoMT-COCO and fine-tuned on Cityscapes.

Model	mIoU
ERFNet	72.50
EoMT-COCO (zero-shot)	55.17
EoMT-Cityscapes	**81.68**
EoMT-Fine-tuned	78.85

3.1. Semantic Segmentation

As mentioned in the Table 1, EoMT-Cityscapes is the strongest model with a score 81.68% between the compared model, and Fine-tuning substantially boost the mIoU. Beside them ERFNet's 72.50 score is competitive for a pixel-based model. Qualitative predictions for these models are shown in Figure 1.

Table 2. Per-class IoU (%) on `Cityscapes validation dataset`: EoMT-COCO (zero-shot with class mapping) versus EoMT-Cityscapes. EoMT-COCO scores 0% on classes with no COCO equivalent.

Class	EoMT-COCO	EoMT-Cityscapes
road	94.13	98.40
sidewalk	64.81	87.36
building	86.77	94.15
wall	46.20	66.07
fence	50.01	65.49
pole	**0.00**	71.04
traffic light	60.79	75.00
traffic sign	**4.74**	82.13
vegetation	88.15	93.02
terrain	56.84	66.60
sky	89.73	95.54
person	70.51	85.38
rider	**0.00**	71.11
car	84.33	95.54
truck	29.37	81.78
bus	71.17	90.32
train	13.63	77.36
motorcycle	63.78	74.35
bicycle	73.34	81.27
Mean	**55.17**	**81.68**

As referenced in the Table 2, some classes have near or 0% IoU score. This happens because there is no equivalent class labels in COCO dataset, and COCO-EoMT can't classify them for this reason. It is not the problem of the model but a label space mismatch. This problem was the motivation for fine-tuning the COCO-EoMT on Cityscapes.

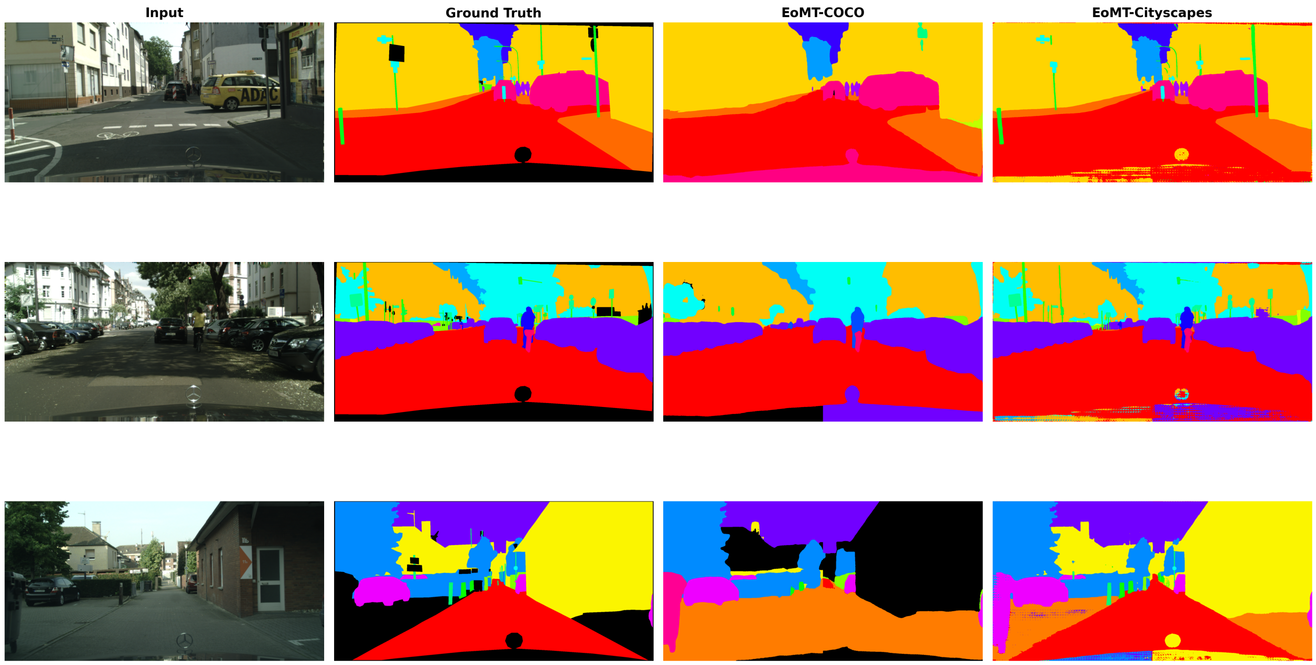

Figure 1. Semantic segmentation results for EoMT-COCO and EoMT-Cityscapes. Left to right: original driving scenes, the ideal color-coded target maps, and the actual pixel-level predictions made by the models.

Fine-tuning lifts mIoU from 55.17 to 78.85, which recovers the most of the gap which is only 3%. This result shows that COCO's DINOv2-backed visual priors working well, the only problem was the classification head, not the features.

3.2. Anomaly Segmentation

Pixel-based vs mask-based. In the Table 3, we see these results from the comparison of ERFNet and EoMT:

- On RoadAnomaly-21 dataset, ERFNet's MaxLogit is 38.31, and EoMT-Cityscapes' MaxEntropy is 77.88. There are 40% difference
- On RoadObstacle-21 dataset, ERFNet's MaxLogit is 4.63 and EoMT-Cityscapes' MSP is 90.11. There are 85% difference
- On RoadAnomaly dataset, ERFNet's MaxLogit 15.58 and EoMT-Cityscapes' MaxEntropy 78.14. There are 63% difference

With these results, we achieved that on most benchmarks, the EoMT's models substantially outperform the ERFNet. Because mask-based classification creates smooth shapes because it looks at the whole object at once. However, pixel-softmax looks at each pixel one by one, that's why it is vulnerable to noise.

Effect of fine-tuning. To see the effect of the fine-tuning, we compare the EoMT-Cityscapes with EoMT-Fine-tuned on AuPRC on the Table 3.

- On RoadAnomaly-21 dataset: MSP score drops from 74.36% to 57.15% (17% change)
- On RoadObstacle-21 dataset: MSP score drops from 90.11% to 73.61% (16% change)
- On RoadAnomaly dataset: MSP score drops from 75.69% to 54.99% (21% change)
- On FS L&F dataset: MSP score increases from 25.42% to 27.11% (2% change)

According to these results, EoMT-COCO recovers closed set mIoU (Table 1) but decrease the anomaly scores compared to training from scratch on Cityscapes. This happens because training the model for a short time makes it hyper-focused on normal objects, and this ruins its ability to spot unusual things.

Comparison of scoring functions. Interesting finding here is that MaxLogit beats MSP and MaxEntropy on ERFNet (38.31 vs 29.09 vs 30.96 on RA-21), but on EoMT they are almost same. This result shows that scoring-method choice depends on the architecture. RbA has a different pattern in here. It achieves highest AuPRC on FS L&F (30.61) and FS Static (56.29) for EoMT-Fine-tuned. However, it also produces high score for FPR95, so it separates the outlier but it is too sensitive so it makes a lot of false alarms.

Qualitative results. Figure 2 shows anomaly score maps for the four scoring methods on representative Road-

Table 3. Anomaly segmentation results across five benchmarks. AuPRC (%) higher is better; FPR95 (%) lower is better. RbA is only applicable to mask-based architectures. Best AuPRC per benchmark in **bold**.

Model	mIoU	Method	RA-21		RO-21		FS L&F		FS Static		RoadAnom	
			AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓
ERFNet	72.50	MSP	29.09	62.57	2.71	65.28	1.75	50.61	7.47	41.84	12.42	82.59
		MaxLogit	38.31	59.36	4.63	48.49	3.30	45.49	9.50	40.32	15.58	73.27
		MaxEntropy	30.96	62.68	3.04	65.96	2.58	50.18	8.84	41.55	12.67	82.76
EoMT-COCO	55.17	MSP	34.44	87.96	1.40	99.99	1.45	96.68	6.19	96.52	20.75	94.01
		MaxLogit	35.20	87.77	1.41	99.99	1.44	96.69	6.24	96.61	21.00	94.08
		MaxEntropy	41.07	80.14	3.67	99.99	1.16	96.45	5.89	97.24	23.40	92.71
		RbA	23.84	64.89	39.36	99.99	0.19	97.13	1.51	99.95	17.85	79.63
EoMT-Cityscapes	81.68	MSP	74.36	33.73	**90.11**	0.55	25.42	14.90	54.35	34.94	75.69	19.31
		MaxLogit	73.99	33.90	90.05	0.58	25.44	14.66	54.44	34.62	75.15	19.06
		MaxEntropy	**77.88**	33.60	90.01	0.68	29.56	14.72	52.51	34.84	**78.14**	18.79
		RbA	69.99	80.06	86.97	99.95	25.97	8.79	56.11	82.68	74.62	20.41
EoMT-Fine-tuned	78.85	MSP	57.15	98.61	73.61	25.09	27.11	26.16	53.41	39.74	54.99	16.22
		MaxLogit	53.69	98.74	73.14	43.77	27.39	25.84	53.61	37.82	55.14	15.68
		MaxEntropy	54.49	98.82	72.62	57.49	28.56	26.33	54.27	39.25	56.22	14.78
		RbA	51.82	99.85	64.22	99.98	**30.61**	26.92	**56.29**	90.34	56.14	13.38

Anomaly21 images.

3.3. Temperature Scaling Study

For temperature scaling we tried $T \in \{0.5, 0.75, 1.0, 1.1, 1.5, 2.0\}$ on MSP, MaxEntropy, and RbA for EoMT-Cityscapes, EoMT-Fine-tuned, and ERFNet. Full per-model results are reported in Tables 4, 5, and 6. Main finding that AuPRC is mostly stable across T.

- In EoMT-Cityscapes (Table 4), MSP AuPRC on RA-21 stays in a 0.3% window, and MaxEntropy stays in a 0.5% window.
- In EoMT-Fine-tuned (Table 5), It follows similar pattern as EoMT-Cityscapes
- In ERFNet (Table 6), MSP on RA-21 has a change range for 3%, but its AuPRC is so low that the gain is not meaningful.

Interesting finding that RbA's AuPRC is stable, but its FPR95 swings with T:

- EoMT-Cityscapes RbA on FS L&F: FPR95=67.27 at T=0.5 vs 8.79 at T=1.0
- EoMT-Cityscapes RbA on RoadAnomaly: FPR95's score increases from 20.41 at T=1.0 to 17.96 at T=2.0.
- EoMT-Fine-tuned RbA on RoadAnomaly: FPR95 = 81.57 at T=0.5 vs 12.09 at T=2.0

As result, we found that T=1.0 is competitive for AuPRC most of all settings. Temperature scaling does not meaningfully improve AuPRC. Temperature scaling can shift the FPR95 results, especially for RbA, but it is not a consistent gain. Post-hoc temperature calibration is not a reliable method for improving anomaly segmentation on these benchmarks.

4. Discussion

Mask-based architectures help anomaly detection. **[TODO: Summarize the central empirical finding.]** **[TODO: Explain why, briefly.]**

Fine-tuning recovers most of the closed-set gap. **[TODO: State the main fine-tuning finding.]** **[TODO: Comment on efficiency.]**

Limitations. **[TODO: Acknowledge limits.]**

Future work. **[TODO: One or two directions.]**

5. Conclusion

[TODO: Recap problem.] **[TODO: Recap method.]** **[TODO: Recap main finding 1 (semantic seg).]** **[TODO: Recap main finding 2 (anomaly seg).]** **[TODO: Closing sentence.]**

Acknowledgments

- We declare that Generative AI tools were not used for writing the project report
- For code development of the project, we used Claude by Anthropic. This tool is used for these reasons:
  - Debugging errors, catching bugs, fixing memory issues during large-scale evaluation
  - Understanding the EoMT and ERFNet code structure
  - Reviewing the formulas, and help with the structring the repository
  - Performance optimization, helping to change data-loading patterns that reduced runtime

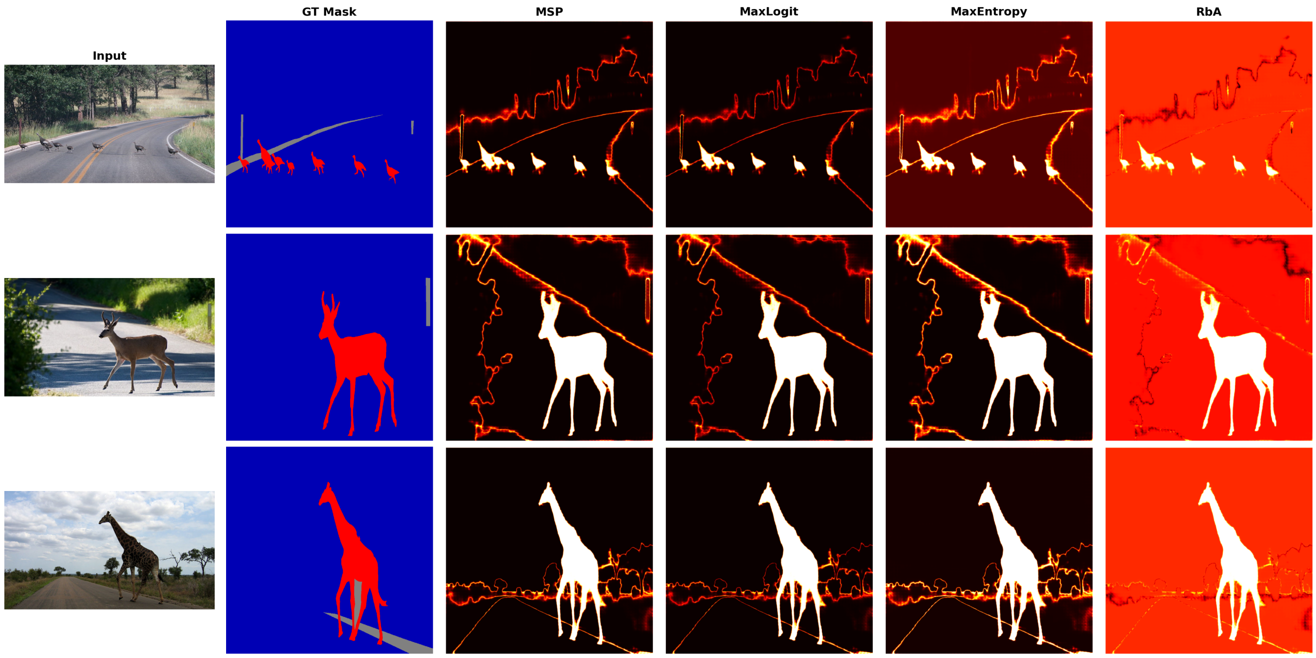

Figure 2. Qualitative comparison of anomaly score maps on RoadAnomaly21. Each row shows an input image alongside score maps produced by MSP, MaxLogit, MaxEntropy, and RbA for EoMT-Cityscapes. Brighter regions indicate higher anomaly score; ground-truth anomaly contours are overlaid in green.

Table 4. Temperature scaling results for **EoMT-Cityscapes**.

		RA-21		RO-21		FS L&F		FS Static		RoadAnom	
Method	mIoU	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓
MSP	81.68	74.36	33.73	90.11	0.55	25.42	14.90	54.35	34.94	75.69	19.31
MSP(t=0.5)		74.21	33.73	90.10	0.55	25.40	14.91	54.33	34.94	75.51	19.37
MSP(t=0.75)		74.29	33.73	90.11	0.55	25.41	14.91	54.34	34.94	75.62	19.33
MSP(t=1.1)		74.40	33.73	90.11	0.55	25.42	14.90	54.35	34.94	75.71	19.30
MSP(t=1.5)		74.50	33.73	90.11	0.55	25.43	14.90	54.35	34.94	75.80	19.29
MSP(t=2.0)		74.55	33.73	90.11	0.55	25.44	14.90	54.36	34.93	75.87	19.28
MaxEntropy	81.68	77.88	33.60	90.01	0.68	29.56	14.72	52.51	34.84	78.14	18.79
MaxEntropy(t=0.5)		77.43	33.68	90.09	0.59	29.55	14.83	53.21	34.91	78.43	19.03
MaxEntropy(t=0.75)		77.71	33.63	90.04	0.64	29.55	14.76	52.86	34.87	78.33	18.88
MaxEntropy(t=1.1)		77.87	33.58	90.01	0.69	29.48	14.71	52.40	34.84	78.08	18.77
MaxEntropy(t=1.5)		77.90	33.55	89.96	0.71	29.28	14.67	51.94	34.79	77.80	18.69
MaxEntropy(t=2.0)		77.91	33.53	89.97	0.72	29.01	14.65	51.46	34.77	77.43	18.63
RbA	81.68	69.99	80.06	86.97	99.95	25.97	8.79	56.11	82.68	74.62	20.41
RbA(t=0.5)		69.57	76.02	73.77	99.95	26.00	67.27	55.85	84.55	74.12	44.02
RbA(t=0.75)		69.79	78.10	84.74	99.95	25.97	16.61	56.16	84.60	74.55	20.37
RbA(t=1.1)		70.00	84.05	87.34	99.95	25.96	9.03	56.08	82.35	74.65	19.76
RbA(t=1.5)		70.39	85.55	87.97	99.95	25.93	9.65	55.98	73.52	75.01	18.34
RbA(t=2.0)		70.83	91.98	88.25	99.94	25.92	10.05	55.91	35.88	75.09	17.96

All generated code was reviewed, tested, and integrated by the authors. All decisions like model choice, training configuration, evaluation protocol, dataset selection and analysis of experimental results were made by the authors.

Table 5. Temperature scaling results for **EoMT-Fine-tuned**.

Method	mIoU	RA-21 AuPRC↑	RA-21 FPR95↓	RO-21 AuPRC↑	RO-21 FPR95↓	FS L&F AuPRC↑	FS L&F FPR95↓	FS Static AuPRC↑	FS Static FPR95↓	RoadAnom AuPRC↑	RoadAnom FPR95↓
MSP	78.85	57.15	98.61	73.61	25.09	27.11	26.16	53.41	39.74	54.99	16.22
MSP(t=0.5)		58.59	98.60	73.69	25.10	27.07	26.22	53.35	39.78	54.96	16.27
MSP(t=0.75)		57.58	98.60	73.63	25.10	27.10	26.17	53.39	39.75	54.98	16.24
MSP(t=1.1)		57.05	98.61	73.60	25.09	27.11	26.16	53.41	39.74	54.99	16.22
MSP(t=1.5)		56.78	98.61	73.58	25.09	27.12	26.16	53.43	39.73	54.99	16.21
MSP(t=2.0)		56.61	98.62	73.57	25.09	27.13	26.15	53.43	39.72	55.00	16.20
MaxEntropy	78.85	54.49	98.82	72.62	57.49	28.56	26.33	54.27	39.25	56.22	14.78
MaxEntropy(t=0.5)		54.10	98.71	73.43	28.83	27.91	26.21	53.87	39.53	55.83	15.28
MaxEntropy(t=0.75)		54.37	98.81	72.96	36.81	28.27	26.26	54.10	39.40	56.08	14.99
MaxEntropy(t=1.1)		54.47	98.82	72.52	45.00	28.60	26.34	54.32	39.19	56.27	14.74
MaxEntropy(t=1.5)		54.58	98.82	72.13	58.44	28.92	26.28	54.43	39.06	56.41	14.61
MaxEntropy(t=2.0)		54.65	98.82	71.69	60.82	29.18	26.24	54.50	38.76	56.33	14.51
RbA	78.85	51.82	99.85	64.22	99.98	30.61	26.92	56.29	90.34	56.14	13.38
RbA(t=0.5)		51.52	99.53	62.10	99.98	31.42	78.97	56.29	89.51	52.39	81.57
RbA(t=0.75)		51.76	99.78	63.36	99.98	30.83	23.06	56.55	90.28	54.60	74.33
RbA(t=1.1)		51.83	99.86	64.43	99.98	30.56	28.24	56.19	90.29	56.30	12.37
RbA(t=1.5)		51.86	99.87	65.73	99.98	30.43	25.75	55.91	90.07	56.51	12.08
RbA(t=2.0)		51.88	99.87	66.12	99.98	30.36	24.49	55.73	89.65	56.57	12.09

Table 6. Temperature scaling results for **ERFNet**. RbA is omitted as it is incompatible with pixel-based architectures.

Method	mIoU	RA-21 AuPRC↑	RA-21 FPR95↓	RO-21 AuPRC↑	RO-21 FPR95↓	FS L&F AuPRC↑	FS L&F FPR95↓	FS Static AuPRC↑	FS Static FPR95↓	RoadAnom AuPRC↑	RoadAnom FPR95↓
MSP	72.50	29.09	62.57	2.71	65.27	1.75	50.62	7.47	41.84	12.42	82.59
MSP(t=0.5)		27.05	62.75	2.42	63.24	1.28	66.73	6.60	43.48	12.19	82.04
MSP(t=0.75)		28.15	62.51	2.57	64.17	1.49	51.77	6.99	42.50	12.32	82.33
MSP(t=1.1)		29.39	62.67	2.76	65.92	1.86	50.21	7.69	41.62	12.46	82.75
MSP(t=1.5)		30.22	63.49	2.93	68.80	2.29	49.12	8.60	41.19	12.57	83.51
MSP(t=2.0)		30.60	65.05	3.00	72.74	2.69	47.92	9.54	41.29	12.64	84.58
MaxEntropy	72.50	30.96	62.67	3.04	65.95	2.58	50.19	8.84	41.55	12.67	82.76
MaxEntropy(t=0.5)		28.58	62.74	2.64	63.73	1.59	52.82	7.04	43.20	12.33	82.26
MaxEntropy(t=0.75)		30.11	62.46	2.87	64.45	2.07	51.46	7.83	42.29	12.51	82.38
MaxEntropy(t=1.1)		31.12	62.92	3.09	66.74	2.76	49.88	9.25	41.34	12.71	82.98
MaxEntropy(t=1.5)		31.07	64.37	3.11	70.77	3.24	48.56	10.64	41.04	12.81	84.06
MaxEntropy(t=2.0)		30.27	67.09	2.97	75.98	3.51	47.10	11.72	41.75	12.79	85.60

References

[1] Hermann Blum, Paul-Edouard Sarlin, Juan Nieto, Roland Siegwart, and Cesar Cadena. Fishyscapes: A benchmark for safe semantic segmentation in autonomous driving. In *ICCV Workshops*, 2019. 1, 3

[2] Robin Chan, Krzysztof Lis, Svenja Uhlemeyer, et al. Segmentmeifyoucan: A benchmark for anomaly segmentation. In *NeurIPS Datasets and Benchmarks*, 2021. 1, 3

[3] Bowen Cheng, Alexander G Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. In *NeurIPS*, 2021. 1

[4] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *CVPR*, 2022. 1, 2

[5] Tom Kerssies and Daan de Geus. An encoder-only approach to mask classification. In *CVPR*, 2025. 1

[6] Nishant Nayal, Yigit Yavuz, João F Henriques, and Daan de Geus. Scaling out-of-distribution detection for real-world settings. In *ECCV*, 2024. 1, 2

[7] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. 1

[8] Eduardo Romera, José M Alvarez, Luis M Bergasa, and Roberto Arroyo. Erfnet: Efficient residual factorized convnet for real-time semantic segmentation. pages 263–272, 2018. 1, 2